

From the two images, the expressions are:

Expression 1

7 + 5 * 3 / 5 ^ 1 + (3 - 2)

Expression 2

8 * (5 ^ 4 + 2) - 6 ^ 2 / (9 * 3)

We use the standard precedence:

Operator	Precedence	Associativity
^	3	Right → Left
*, /	2	Left → Right
+, -	1	Left → Right
(	Special	—

Example 1

Infix

7 + 5 * 3 / 5 ^ 1 + (3 - 2)

We scan **left to right**.

Infix → Postfix

Step	Token	Action	Stack	Postfix Output
1	7	Operand → output	—	7
2	+	Push +	+	7
3	5	Operand → output	+	75
4	*	* has higher precedence than + → push	+ *	75

Step	Token	Action	Stack	Postfix Output
5	3	Operand → output	+ *	753
6	/	* has equal precedence → pop *, then push /	+ /	753*
7	5	Operand → output	+ /	753*5
8	^	^ has higher precedence → push	+ / ^	753*5
9	1	Operand → output	+ / ^	753*51
10	+	Pop ^, /, +; then push +	+	753*51^/+
11	(	Push (	+ (	753*51^/+
12	3	Operand → output	+ (	753*51^/+3
13	-	Push -	+ (-	753*51^/+3
14	2	Operand → output	+ (-	753*51^/+32
15	)	Pop -; discard (	+	753*51^/+32-
16	END	Pop +	—	753*51^/+32-+

Final Postfix

7 5 3 * 5 1 ^ / + 3 2 - +

or without spaces:

753*51^/+32-+

Check the conversion

The original expression means:

$7 + ((5 * 3) / (5 ^ 1)) + (3 - 2)$

Postfix:

7 5 3 * 5 1 ^ / + 3 2 - +

Correct.

Example 2

Infix

8 * (5 ^ 4 + 2) - 6 ^ 2 / (9 * 3)

Again, scan from **left to right**.

Infix → Postfix

Step	Token	Action	Stack	Postfix Output
1	8	Operand → output	—	8
2	*	Push *	*	8
3	(	Push (	* (	8
4	5	Operand → output	* (	85
5	^	Push ^	* (^	85
6	4	Operand → output	* (^	854
7	+	Pop ^; push +	* (+	854^
8	2	Operand → output	* (+	854^2
9	)	Pop +; discard (	*	854^2+
10	-	* has higher precedence → pop *; push -	-	854^2+*
11	6	Operand → output	-	854^2+*6
12	^	Push ^	- ^	854^2+*6
13	2	Operand → output	- ^	854^2+*62

Step	Token	Action	Stack	Postfix Output
14	/	$\wedge$ has higher precedence $\rightarrow$ pop $\wedge$; push /	- /	854^2+*62^
15	(	Push (	- / (	854^2+*62^
16	9	Operand $\rightarrow$ output	- / (	854^2+*62^9
17	*	Push *	- / (*	854^2+*62^9
18	3	Operand $\rightarrow$ output	- / (*	854^2+*62^93
19	)	Pop *; discard (	- /	854^2+*62^93*
20	END	Pop /	-	854^2+*62^93*/
21	END	Pop -	—	854^2+*62^93*/-

Final Postfix

8 5 4 ^ 2 + * 6 2 ^ 9 3 * / -

or without spaces:

854^2+*62^93*/-

Verification of Example 2

The original expression:

8 * (5^4 + 2) - 6^2 / (9*3)

can be grouped according to precedence:

[8 * ((5^4) + 2)] - [(6^2) / (9*3)]

Now convert each part.

First part

$$8 * (5^4 + 2)$$

Postfix:

$$8 \ 5 \ 4 \ ^ \ 2 \ + \ *$$

Second part

$$6^2 / (9 * 3)$$

Postfix:

$$6 \ 2 \ ^ \ 9 \ 3 \ * \ /$$

Combine with **-**

$$8 \ 5 \ 4 \ ^ \ 2 \ + \ * \ 6 \ 2 \ ^ \ 9 \ 3 \ * \ / \ -$$

Therefore:

$$854^2 + * 62^9 3 * / -$$

is correct.

Final Answers

Expression	Postfix
$7 + 5 * 3 / 5^1 + (3 - 2)$	$753 * 51 ^ / + 32 - +$
$8 * (5^4 + 2) - 6^2 / (9 * 3)$	$854^2 + * 62^9 3 * / -$

With spaces

1. $7 \ 5 \ 3 \ * \ 5 \ 1 \ ^ \ / \ + \ 3 \ 2 \ - \ +$

2. $8 \ 5 \ 4 \ ^ \ 2 \ + \ * \ 6 \ 2 \ ^ \ 9 \ 3 \ * \ / \ -$

Key point in both examples: whenever an operator arrives, compare its precedence with the operator at the top of the stack. Higher-precedence operators already on the stack are popped before pushing the new operator. Parentheses temporarily stop this popping.

